

The Impact of Digital Learning Platforms on Student Engagement and Academic Performance in Higher Education

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Purpose of the Study

This quantitative study aims to examine the relationships among digital learning platform use, student engagement, and academic achievement among undergraduate students in higher education. Knowing how well digital technologies improve student results is crucial as they continue to influence the way that education is delivered. Despite the widespread usage of digital platforms like learning management systems (LMS) and online collaboration tools, it is still unknown how they actually affect quantifiable learning outcomes.

Recent studies suggest that the integration of digital learning technologies can enhance student engagement and academic achievement; however, the extent of these benefits depends on how effectively such technologies are implemented and utilized within educational settings. Research has shown that students' digital literacy plays an important role in maximizing the educational value of digital platforms. Students with stronger digital literacy skills are more likely to participate actively in online learning environments and achieve better academic outcomes (Getenet et al., 2024; Zakir et al., 2025). Therefore, this study seeks to determine whether greater use of digital learning platforms is associated with higher levels of student engagement and academic achievement and whether digital literacy moderates these relationships among undergraduate students in higher education.

Research Questions

1. What connection exists between student engagement and the use of digital learning platforms?
2. What impact does student involvement have on academic achievement?

3. Does the use of digital platforms and student engagement have a moderating effect on digital literacy?

Significance of the Study

The quantitative evidence this study provides about the efficacy of digital learning platforms in higher education makes it noteworthy. Determining whether educational technologies result in quantifiable gains in student engagement and academic achievement is crucial as educational institutions continue to invest in them.

Examining these correlations is important, according to a number of peer-reviewed quantitative research. For instance, studies using data from learning management systems (LMS) have discovered that student interaction patterns are a strong predictor of academic achievement results (Xu et al., 2024). Similarly, research in online higher education has shown a positive correlation between academic accomplishment and the amount of time spent on digital platforms and the frequency of involvement (Frontiers in Education, 2022).

Furthermore, Teng and Wang (2021) discovered that educational technology tools had a substantial impact on behavioral, emotional, and cognitive aspects of student involvement. According to another quantitative study by McKenna et al. (2024), quantitative markers can be used to quantify student participation at various stages of higher education. Together, this research demonstrate how crucial digital platforms are in determining learning outcomes. They also highlight a lack of knowledge regarding the relationship between platform usage, engagement, and academic achievement within a particular model. By combining these characteristics into a single quantitative framework, this study fills that gap and advances educational technology theory and practice.

Research Design

This study will employ a quantitative correlational research design to examine the relationships among digital learning platform use, student engagement, digital literacy, and academic achievement among undergraduate students. Student engagement and digital literacy will be measured using validated survey instruments, while digital learning platform use will be assessed through learning management system (LMS) analytics, including login frequency, time spent on the platform, and activity completion rates. Academic achievement will be determined using students' course grades or examination scores. Pearson correlation analysis and multiple regression techniques will be used to determine whether digital learning platform use and student engagement are significantly associated with academic achievement and to examine whether digital literacy moderates these relationships.

Participants

Undergraduate students taking courses that make use of digital learning tools will be among the participants. To provide adequate statistical power, a sample size of between 150–300 students will be the goal.

Data Collection Instruments

Data will be collected through a structured questionnaire administered to undergraduate students enrolled in courses that utilize digital learning platforms. The questionnaire will consist of three sections: (1) items measuring students' use of digital learning platforms, adapted from recent studies on technology use in higher education; (2) a validated student engagement scale assessing behavioral, emotional, and cognitive engagement; and (3) a digital literacy scale that measures

students' perceived competence and confidence in using digital technologies.

Variables

- Use of digital learning platforms is an independent variable.
- Academic achievement is the dependent variable.
- Student engagement is a mediating variable.
- Digital literacy is a moderating variable.

Analysis of Data

- The analysis of the data will be done using:
- Descriptive statistics Pearson correlation analysis
- Analysis of multiple regression
- If necessary, structural equation modeling (SEM)

Definitions

- Digital learning platforms are online tools like learning management systems (LMS) that are used to distribute course materials and promote engagement.
- The degree of student participation in educational activities, encompassing behavioral, emotional, and cognitive aspects (Fredricks et al., 2004).

- Academic Performance: Quantifiable markers of academic success, like test scores and grades.
- Digital literacy is the capacity to learn and solve problems using digital technology (Ng, 2012).
- A mediating variable is one that clarifies the connection between an independent and dependent variable.
- A variable that influences the direction or strength of the link between variables is known as a moderating variable.

Limitations

This study has some drawbacks. First, response bias may be introduced when using self-reported survey data to gauge student participation (Fredricks et al., 2004). Second, the study's focus on undergraduate students may limit the findings' applicability to other demographics. Third, causal links cannot be established because of the correlational design. Furthermore, although they are not entirely under control, outside variables including student motivation, course design, and instructor efficacy may affect the outcomes (Schindler et al., 2017).

Delimitation's

Only undergraduate students enrolled in courses that make use of digital learning platforms are included in this study. It ignores other possible implications like socioeconomic background or past academic success in favor of concentrating on particular factors like platform usage, engagement, academic performance, and digital literacy. Furthermore, the study only employs quantitative techniques and excludes qualitative information like open-ended questions or interviews.

Assumptions

This research is predicated on a number of hypotheses. It is predicated on respondents giving truthful and precise answers to survey questions. Additionally, it presupposes the validity and reliability of the tools used to gauge student involvement and digital literacy (Ng, 2012). Furthermore, it is believed that platform utilization statistics fairly depict how students engage with digital resources. Lastly, the study makes the assumption that statistical techniques are suitable for examining the correlations between variables.

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